

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458991.7256 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1378 +/- 0.0058
Transit depth (Rp/Rs)^2	1.9 +/- 0.16 [%]
Orbital Inclination (inc)	87.13 +/- 0.46
Airmass coefficient 1 (a1)	1.3655 +/- 0.0068
Airmass coefficient 2 (a2)	-0.286 +/- 0.03
Scatter in the residuals of the lightcurve fit is	0.5 %
Best Comparison Star	None
Optimal Aperture	4.61
Optimal Annulus	12.92
Transit Duration (day)	0.0895 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1378 $\pm$ 0.0058 vs 0.1063 (deviation 4.86 σ)
Duration fit vs published	0.0895 d vs 0.0867 d (deviation 0.62 σ)
Mid-time O-C	-4.0 min
Depth SNR	21.3
Residual scatter	0.5 %

Light curve

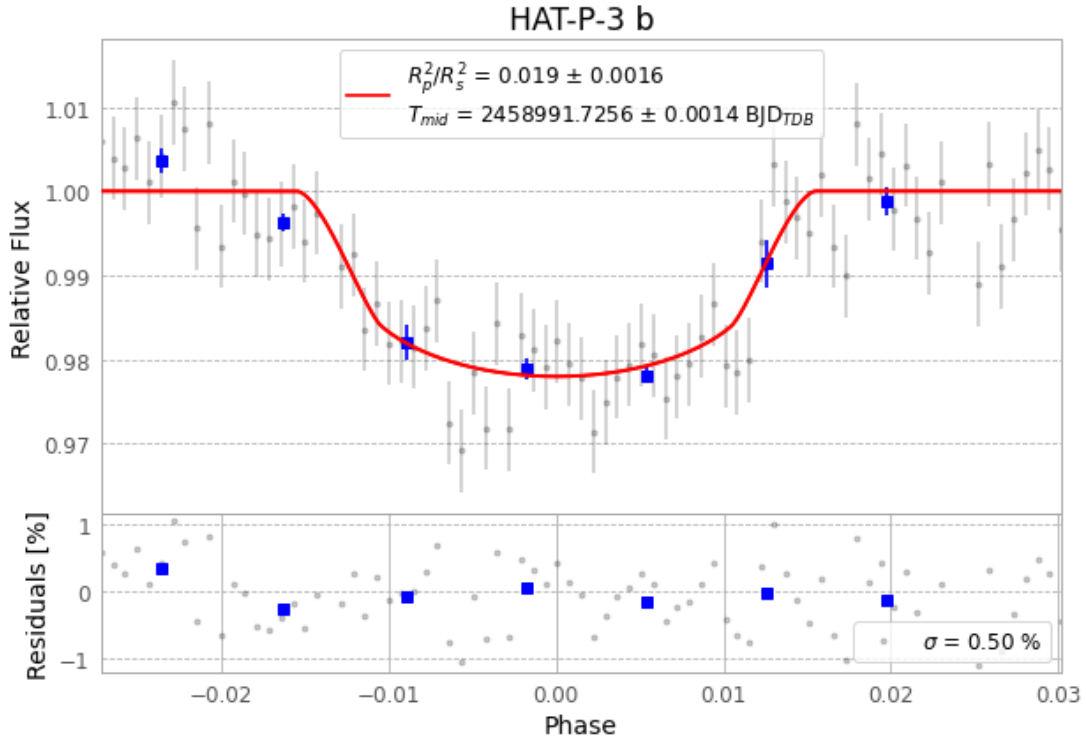


Figure 1 — FinalLightCurve_HAT-P-3 b_2020-05-21.png (SHA-256 eedfa66dd1153fdb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [488, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a5382aee2320aa75...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), HATP-3200522032713.FITS → HATP-3200522072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-200522.log (c13827d4746c...), FinalLightCurve_HAT-P-3 b_2020-05-21.png (eedfa66dd115...), FinalLightCurve_HAT-P-3 b_2020-05-21.csv (9fec9985b329...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 18:05Z.